NACETEM POLICY BRIEF
2025 Number 1

ENHANCING KNOWLEDGE TRANSFER SYSTEMS TO BOOST AGRICULTURAL PRODUCTIVITY AND FOOD SECURITY IN NIGERIA

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Background

Knowledge transfer involves the systematic exchange of information, skills, and technologies among relevant stakeholders in the agricultural sector including farmers, research institutions, extension agents, policymakers, and NGOs/CSOs, among others. Essentially, knowledge transfer ensures a closely linked interaction among critical stakeholders in the sector. In this regard, farmers benefit from up-to-date agricultural research outputs/innovations; simultaneously, agricultural researchers and policymakers make informed decisions, based on the feedback from farmers. Consequently, knowledge transfer is vital for improving agricultural productivity, reducing post-harvest losses, and building resilience to climate change. Knowledge transfer also fosters collaboration, enabling a holistic stakeholder approach to addressing complex challenges confronting the agricultural sector in Nigeria.

Despite ongoing agricultural research and development efforts in Nigeria, the flow of knowledge from research institutions to farmers remains fragmented and inefficient (Ameh et al., 2023). Weak linkages among stakeholders—research bodies, extension services, agribusinesses, and farmers—limit the adoption of innovations crucial for improving food production and resilience. This gap in knowledge transfer mechanisms hampers the country's ability to achieve sustainable food security. The National Agricultural Technology and Innovation Policy (NATIP) aims to address these challenges by promoting sustainable practices, innovation, and productivity. However, the success of NATIP depends on effective knowledge transfer—the process of sharing, disseminating, and applying knowledge among stakeholders in the agricultural sector.

Current State and Challenges

The current state of knowledge dissemination in Nigerian agriculture reflects both progress and persistent challenges. Agricultural Development Programs (ADPs) serve as the main public extension service providers, employing around 7,000 agents, of which 28% are female (Agwu et al., 2023). However, the agent-to-farmer ratio is critically low, meeting only about 25% of national extension needs, thus limiting reach and impact, especially in rural communities. While digital technology adoption is growing—evidenced by 54% of farmers in Ebonyi State having access to mobile phones—usage of advanced tools like E-wallet apps or social media remains low. Furthermore, only 3% of farmers reported using the E-wallet app for fertilizer information, and just 1.3% used Instagram for agricultural purposes (Onomu et al., 2024). Nonetheless, ICT-based interventions have demonstrated positive outcomes, as seen in Abuja's Kwali Area Council, where 85% of farmers reported improvements due to digital tools.



Traditional media, particularly radio, continues to play a major role in knowledge transfer, though it faces limitations related to language, coverage, and content relevance (Ashindo and Danjuma, 2021). Simultaneously, indigenous knowledge remains vital in rural areas, with many farmers relying on traditional methods. Integrating these systems with modern agricultural practices is essential for sustainability and food security.

Despite these efforts, challenges such as limited digital access, low extension staffing (with ratios as high as 1:10,000), and cultural resistance to new practices hamper progress. In response, the Nigerian government has introduced policy reforms and digital platforms to bridge these gaps. Notable among these is the 2023 National Agricultural Extension Policy and the 2025 launch of the National Electronic Extension Platform (NEEP), alongside increased recruitment of agricultural scientists by Agricultural Research Council of Nigeria (ARCN). Tools like Farmonaut (Farmonaut is a digital agriculture solutions company that uses satellite technology to provide farmers with tools for precision agriculture) also reflect the growing shift toward data-driven, technology-enabled farming. These developments underscore a national commitment to strengthening knowledge dissemination for improved agricultural productivity and food security.

Proposed Strategies

Promoting collaboration and leveraging technology to improve knowledge transfer mechanisms among stakeholders is essential for achieving food security in Nigeria. This requires a coordinated, multi-pronged approach that brings together institutions, farmers, researchers, and the private sector in a more structured and impactful way.

One of the key strategies is fostering multi-stakeholder platforms and public-private partnerships (PPPs). Effective knowledge transfer thrives on strong collaboration among government bodies, academic institutions, research organizations, agribusinesses, and farmers' groups. Countries like the Netherlands have adopted the "Golden Triangle" model, where government, research institutions, and businesses jointly drive agricultural innovation. Similarly, in the United States, the Cooperative Extension System connects universities with farmers to disseminate research-based information. Nigeria can adopt similar models by strengthening existing institutions like the ARCN and incorporating structured PPPs. Establishing regional knowledge hubs that link universities such as Ahmadu Bello University and the University of Ibadan with research institutes like International Institute of Tropical Agriculture (IITA) and innovative agritech startups can help co-develop context-specific agricultural solutions.

Another vital strategy is the use of digital technologies to facilitate knowledge sharing. Emerging technologies such as artificial intelligence, the Internet of Things (IoT), blockchain, and mobile-based advisory services can enable real-time communication and decision-making. Countries like India and Israel have successfully adopted such tools to enhance extension services. India's e-Choupal and Digital Green, for

example, provide farmers with tailored advice and market updates using digital platforms. In Israel, real-time satellite data and AI are used to guide irrigation and crop management. Nigeria can follow suit by deploying AI-based advisory systems that use localized data on soil, weather, and crops. Scaling up existing platforms such as FarmCrowdy, ThriveAgric, and AgroMall—with support from the Ministry of Agriculture—can accelerate this transformation. In addition, the government should collaborate with telecommunications providers to offer subsidized data plans for agriculture-related apps and develop USSD (Unstructured Supplementary Service Data) and IVR ((Interactive Voice Response) services in local languages to reach low-literate farmers.

Capacity building and training are also critical components of effective knowledge transfer. Providing farmers with hands-on training and digital tools enhances their ability to access new techniques, funding opportunities, and market information. China's model of Farmer Field Schools (FFS) and Brazil's Lead Farmer approach demonstrate the



benefits of practical, peer-led learning (China's FFS is a participatory, hands-on learning process where farmers collectively learn through experimentation and observation, while the Brazil's Lead Farmer approach relies on a designated individual who disseminates information and practices to their peers). Nigeria should strengthen and expand its existing FFS programs through institutions like the National Agricultural Extension and Research Liaison Services (NAERLS). Training a network of tech-savvy lead farmers can promote peer learning and ensure grassroots dissemination of innovations. Moreover, workshops, webinars, and field demonstrations on topics such as climate-smart agriculture and sustainable farming practices can boost adoption of modern techniques.

Improving knowledge-sharing networks is another important step. Establishing co-creation hubs where farmers, researchers, and agribusiness stakeholders can collaborate will foster joint innovation and problem-solving. These hubs can also serve as spaces for experience sharing, experimentation, and continuous learning. Additionally, creating open-access knowledge platforms and digital libraries that compile research outputs, extension materials, and market intelligence will democratize access to agricultural knowledge. Models such as the European Union's Open Data Portal for Agriculture and United Sates agency for International Development's (USAID) Feed the Future initiative serve as valuable examples. Nigeria should consider launching a National Open Agriculture Portal—possibly under the National Information Technology Development Agency (NITDA) or the Federal Ministry of Agriculture and Rural Development (FMARD)—to host research outputs, weather data, pest alerts, and other key resources. Making outputs from Nigerian research institutes such as IITA and National Cereals Research Institute (NCRI) freely accessible and user-friendly will further support informed decision-making.

Inclusion of youth and women in agritech development is another crucial element. Engaging these often-marginalized groups can unlock new sources of innovation and sustainability. Countries like Rwanda and Kenya have established agritech hubs that train youth and women as data agents and technology entrepreneurs. The United States, through NGO partnerships, targets women in its extension programs. In Nigeria, initiatives such as the Youth Empowerment in Agriculture Programme (YEAP) should be expanded to offer agritech innovation grants. Targeted digital literacy and entrepreneurship training should also be provided to women's cooperatives and members of the National Youth Service Corps (NYSC) to encourage their active participation in the agricultural sector.

Finally, policy and infrastructure support are indispensable. The government must invest in expanding rural electrification, internet connectivity and digital inclusion. Financial incentives, grants, and regulatory frameworks should be put in place to support startups that are developing innovative agricultural solutions. Establishing a National Food Systems Intelligence Unit to track trends, monitor outcomes, and support evidence-based policymaking would provide a strong institutional backbone for long-

term planning and responsiveness. Summarily, by integrating collaborative platforms, digital technologies, inclusive training programs, and supportive policies, Nigeria can significantly enhance knowledge transfer mechanisms in its agricultural sector. These efforts will not only empower farmers and improve productivity but also contribute to building a more resilient and food-secure nation.

Policy Recommendations

(i) Implement a nationwide ICT-enabled extension strategy that blends traditional field extension with digital platforms such as mobile apps, USSD/IVR services, and AI-powered advisory tools in local languages. This will leverage the growing digital ecosystem within Nigeria's agricultural extension system to close the knowledge gap, particularly in underserved rural areas, and improves real-time dissemination of innovations.

(ii) Develop policy incentives (e.g., tax breaks, grants, joint research funding) to encourage partnerships between public research institutions and private agritech firms for the co-development and commercialization of agricultural technologies. This will align public R&D outputs with market-ready solutions and increases private sector engagement.

(iii) Design and implement real targeted programs to build knowledge-sharing capacities among youth, women, and local farmer cooperatives through digital literacy, entrepreneurship training, and innovation grants. Expanding human capital within the agricultural extension system will enhance bottom-up innovation and ensures that all groups actively participate in and benefit from knowledge flows.

(iv) Mandate the creation of a centralized, open-access digital platform under NITDA or FMARD that aggregates research outputs, weather data, market information, and policy briefs from all agricultural actors. This will facilitate transparency, accessibility, and evidence-based decision-making across the innovation system, while fostering stronger linkages between knowledge producers and users.

Conclusion and Call to Action

Now is the critical moment to act on promoting technology transfer and ensuring that research outputs reach Nigerian farmers. Delays will only worsen food insecurity, especially as the population is projected to reach 239 million by 2026, with youth accounting for nearly 70%. Rapid implementation of the recommendations in this brief is essential to unlock the full potential of Nigeria's agricultural sector, achieve sustainable food security, reduce youth unemployment, and curb social vulnerabilities. The decisions taken today will shape the nation's future—decisive action is required now to secure food systems and safeguard economic stability

Enhancing knowledge transfers to Nigerian farmers will go a long way in lifting Nigeria out of its present state of food insecurity (21.3% prevalence in 2022, STATISTA 2025), substantially reducing volume of food imports, increasing farmers' productivities, building resilience, and reducing consumer price index (CPI) of food in the country. Agriculture has the potential to become the biggest contributor to the country's GDP if the right steps are taken on the strategies identified and recommendations made. Such steps will also require political will, proactive support systems, as well as a well-coordinated and inclusive sectoral innovation system that actively connects research institutions, extension agents, agribusinesses, and farmers. By institutionalizing collaborative platforms, scaling digital extension services, fostering inclusive capacity-building, and investing in open-access knowledge systems, Nigeria can transform its fragmented knowledge landscape into a vibrant, innovation-driven ecosystem that ensures innovations reach those who need them most—farmers on the frontlines of food production.

References

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